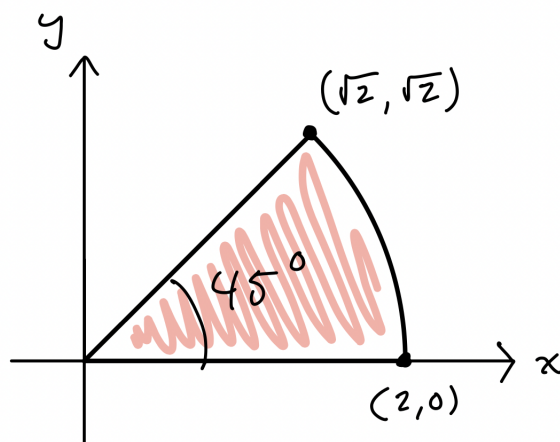


**Problem 1. An Integral in the Plane.** Consider the function  $f(x, y) = x$ . Let  $D$  be the region that is inside the circle  $x^2 + y^2 = 4$ , above the line  $y = 0$  and below the line  $y = x$ .

- Draw the region. [Hint: It looks like 1/8 of a pie.]
- Compute the integral  $\iint_D f(x, y) dx dy$  by converting to polar coordinates.
- Compute the integral  $\iint_D f(x, y) dx dy$  in Cartesian coordinates by cutting the region  $D$  into two pieces  $D_1$  and  $D_2$  separated by the line  $x = \sqrt{2}$ . Check that your answers from parts (a) and (b) are the same.

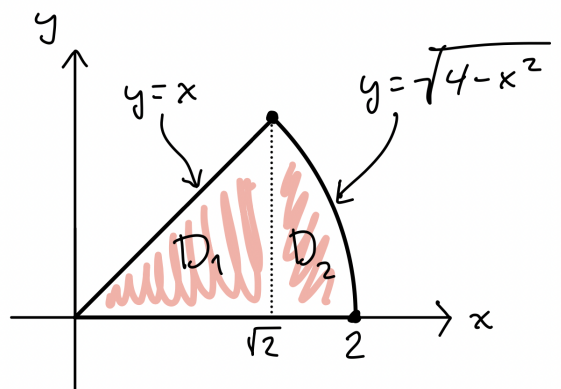
(a):



(b): The region  $D$  is parametrized in polar coordinates by  $0 \leq r \leq 2$  and  $0 \leq \theta \leq \pi/4$ . Hence

$$\begin{aligned}
 \iint_D x dx dy &= \iint_D x r dr d\theta \\
 &= \iint_D r \cos \theta r dr d\theta \\
 &= \iint_D r^2 \cos \theta dr d\theta \\
 &= \int_0^2 r^2 dr \cdot \int_0^{\pi/4} \cos \theta d\theta && \text{(separable)} \\
 &= \left[ \frac{r^3}{3} \right]_0^2 \cdot [\sin \theta]_0^{\pi/4} \\
 &= \left( \frac{2^3}{3} - \frac{0^3}{3} \right) \cdot (\sin(\pi/4) - \sin 0) \\
 &= \frac{8}{3} \cdot \frac{\sqrt{2}}{2} \\
 &= \frac{4\sqrt{2}}{3}.
 \end{aligned}$$

(c): We divide the region  $D$  into  $D_1$  and  $D_2$  as follows:



The region  $D_1$  is parametrized by  $0 \leq x \leq \sqrt{2}$  and  $0 \leq y \leq x$ , so that

$$\begin{aligned}
 \iint_{D_1} x \, dx \, dy &= \int_0^{\sqrt{2}} \left( \int_0^x x \, dy \right) dx \\
 &= \int_0^{\sqrt{2}} [xy]_{y=0}^{y=x} dx \\
 &= \int_0^{\sqrt{2}} x^2 dx \\
 &= \left[ \frac{x^3}{3} \right]_0^{\sqrt{2}} \\
 &= \frac{2\sqrt{2}}{3}.
 \end{aligned}$$

The region  $D_2$  is parametrized by  $\sqrt{2} \leq x \leq 2$  and  $0 \leq y \leq \sqrt{4-x^2}$ , so that

$$\begin{aligned}
 \iint_{D_2} x \, dx \, dy &= \int_{\sqrt{2}}^2 \left( \int_0^{\sqrt{4-x^2}} x \, dy \right) dx \\
 &= \int_{\sqrt{2}}^2 [xy]_{y=0}^{y=\sqrt{4-x^2}} dx \\
 &= \int_{x=\sqrt{2}}^{x=2} x \sqrt{4-x^2} dx \\
 &= \int_{u=0}^{u=2} -\frac{1}{2} \sqrt{u} \, du \quad (u = 4 - x^2, du = -2x dx) \\
 &= \int_0^2 \frac{1}{2} \sqrt{u} \, du \\
 &= \left[ \frac{u^{3/2}}{3} \right]_0^2 \\
 &= \frac{2\sqrt{2}}{3}.
 \end{aligned}$$

We conclude that

$$\iint_D x \, dx \, dy = \iint_{D_1} x \, dx \, dy + \iint_{D_2} x \, dx \, dy = \frac{2\sqrt{2}}{3} + \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{3},$$

which matches our answer from part (b).

[Remark: This problem illustrates the benefit of polar coordinates.]

**Problem 2. Center of Mass.** Let  $D$  be the same region as in Problem 1. Think of this as a thin metal plate with a constant density of 1 unit of mass per unit of area. Compute the following using polar coordinates.

- (a) Compute the total mass  $\iint_D 1 \, dxdy$ .
- (b) Compute the moment about the  $y$  axis:  $\iint_D x \, dxdy$ .
- (c) Compute the moment about the  $x$  axis:  $\iint_D y \, dxdy$ .
- (d) Find the center of mass.

(a): As in Problem 1, we can parametrize this region in polar coordinates as  $0 \leq r \leq 2$  and  $0 \leq \theta \leq \pi/4$ . The area is

$$\begin{aligned} \text{area}(D) &= \iint_D 1 \, dxdy \\ &= \iint_D 1 \, r dr d\theta \\ &= \int_0^2 r \, dr \cdot \int_0^{\pi/4} 1 \, d\theta && \text{(separable)} \\ &= \left[ \frac{r^2}{2} \right]_0^2 \cdot [\theta]_0^{\pi/4} \\ &= \left( \frac{2^2}{2} - \frac{0^2}{2} \right) \cdot \left( \frac{\pi}{4} - 0 \right) \\ &= \frac{\pi}{2}. \end{aligned}$$

Indeed, our pie slice is  $1/8$  of a circle of radius 2, which has area  $\pi(2)^2 = 4\pi$ .

(b): We already computed this in Problem 1(b). The answer is  $4\sqrt{2}/3$ .

(c): Here we have

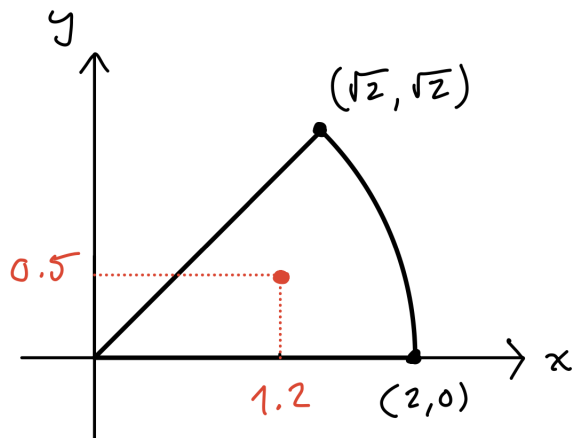
$$\begin{aligned} \iint_D y \, dxdy &= \iint_D y \, r dr d\theta \\ &= \iint_D r \sin \theta \, r dr d\theta \\ &= \iint_D r^2 \sin \theta \, dr d\theta \\ &= \int_0^2 r^2 \, dr \cdot \int_0^{\pi/4} \sin \theta \, d\theta && \text{(separable)} \\ &= \left[ \frac{r^3}{3} \right]_0^2 \cdot [-\cos \theta]_0^{\pi/4} \\ &= \left( \frac{2^3}{3} - \frac{0^3}{3} \right) \cdot (-\cos(\pi/4) + \cos 0) \\ &= \frac{8}{3} \cdot \left( -\frac{\sqrt{2}}{2} + 1 \right) \end{aligned}$$

$$= \frac{4(2 - \sqrt{2})}{3}.$$

(d): The center of mass is  $(\bar{x}, \bar{y})$  where

$$\bar{x} = \frac{\iint_D x}{\iint_D 1} = \frac{4\sqrt{2}/3}{\pi/2} = \frac{8\sqrt{2}}{3\pi} \quad \text{and} \quad \bar{y} = \frac{\iint_D y}{\iint_D 1} = \frac{4(2 - \sqrt{2})/3}{\pi/2} = \frac{8(2 - \sqrt{2})}{3\pi}.$$

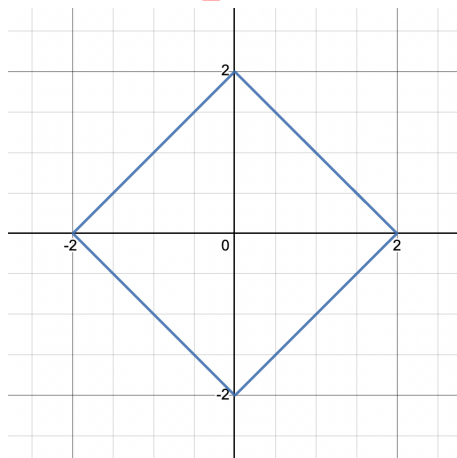
My computer says that  $(\bar{x}, \bar{y}) \approx (1.2, 0.5)$ . Here is a picture:



**Problem 3. Change of Coordinates.** Consider the function  $f(x, y) = x^2 + y^2$ . Let  $D$  be the square-shaped region in the  $x, y$ -plane bounded by the four lines  $x + y = \pm 2$  and  $x - y = \pm 2$ .

- Draw the region.
- Consider the change of variables  $x = u + v$  and  $y = u - v$ . Compute the area stretch factor (i.e., the absolute value of the determinant of the Jacobian matrix.)
- Compute the integral  $\iint_D (x^2 + y^2) dx dy$  by converting to  $u, v$ -coordinates. [Hint: The region  $D$  in the  $u, v$ -plane is parametrized by  $-1 \leq u \leq 1$  and  $-1 \leq v \leq 1$ .]

(a): Desmos produced the following picture:<sup>1</sup>



<sup>1</sup><https://www.desmos.com/calculator/jaxo0awiao>

(b): This region is a little bit difficult to parametrize in Cartesian coordinates, so we change coordinates to  $x = u + v$  and  $y = u - v$ . The area stretch factor is

$$\left| \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix} \right| = \left| \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right| = |(1)(-1) - (1)(1)| = |-2| = 2.$$

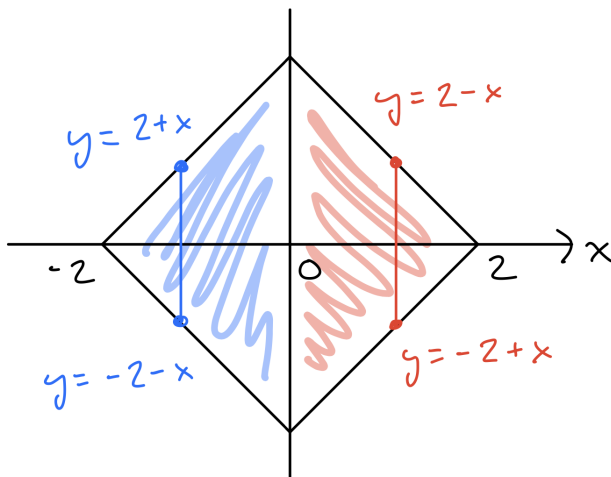
(c): From part (b) we know that  $dx dy = 2 du dv$ . We also have

$$x^2 + y^2 = (u + v)^2 + (u - v)^2 = u^2 + 2uv + v^2 + u^2 - 2uv + v^2 = 2(u^2 + v^2).$$

Furthermore, we note that  $x + y = (u + v) + (u - v) = 2u$  and  $x - y = (u + v) - (u - v) = 2v$ , so the region defined by  $-2 \leq x + y \leq 2$  and  $-2 \leq x - y \leq 2$  becomes  $-2 \leq 2u \leq 2$  and  $-2 \leq 2v \leq 2$ , i.e.,  $-1 \leq u \leq 1$  and  $-1 \leq v \leq 1$ . Thus we have

$$\begin{aligned} \iint_D (x^2 + y^2) dx dy &= \iint_D 2(u^2 + v^2) 2 du dv \\ &= 4 \int_{-1}^1 \left( \int_{-1}^1 (u^2 + v^2) du \right) dv \\ &= 4 \int_{-1}^1 \left[ \frac{u^3}{3} + v^2 u \right]_{u=-1}^{u=1} dv \\ &= 4 \int_{-1}^1 \left( \frac{2}{3} + 2v^2 \right) dv \\ &= 4 \left[ \frac{2}{3} v + 2 \frac{v^3}{3} \right]_{-1}^1 \\ &= 4 \left( \frac{4}{3} + \frac{4}{3} \right) \\ &= \frac{32}{3}. \end{aligned}$$

Remark: One can check that this answer is correct by doing the more difficult computation in Cartesian coordinates. The sideways square  $D$  breaks into two triangles  $D_1$  and  $D_2$  where  $D_1$  is parametrized by  $-2 \leq x \leq 0$  and  $-2 - x \leq y \leq 2 + x$  and where  $D_2$  is parametrized by  $0 \leq x \leq 2$  and  $-2 + x \leq y \leq 2 - x$ :

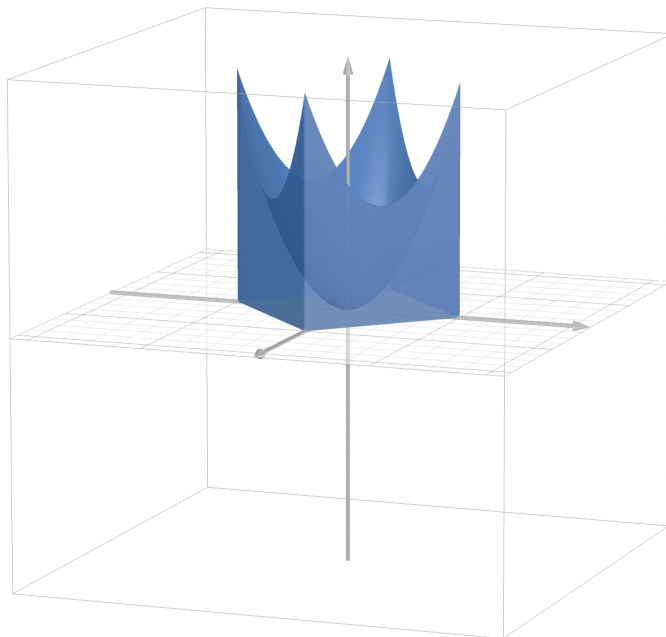


Then the integral over  $D$  is the sum of the integrals over  $D_1$  and  $D_2$ . The integral over  $D_1$  is

$$\begin{aligned}\iint_{D_1} (x^2 + y^2) dx dy &= \int_{-2}^0 \left( \int_{-2-x}^{2+x} (x^2 + y^2) dy \right) dx \\ &= \int_{-2}^0 \left[ x^2 y + \frac{y^3}{3} \right]_{y=-2-x}^{y=2+x} dx \\ &= \int_{-2}^0 \left( x^2(2+x) + \frac{(2+x)^3}{3} - x^2(-2-x) - \frac{(-2-x)^3}{3} \right) dx,\end{aligned}$$

and **we do not want to compute this by hand**. I put it into my computer and it gave the answer  $16/3$ . The integral over  $D_2$  is also  $16/3$ , so their sum is  $32/3$ , as expected.

Remark: We can think of the number  $32/3$  as the **mass** of a thin plate  $D$  with density function  $x^2 + y^2$ , so the corners of  $D$  are heavier than the center. Or we can think of  $32/3$  as the **volume** between the square  $D$  in the  $x, y$ -plane and the surface  $z = x^2 + y^2$  in  $x, y, z$ -space, which looks like a square-shaped crown:<sup>2</sup>



**Problem 4. Integration Over a Rectangular Box.** Let  $B$  be the rectangular box parametrized by  $0 \leq x \leq 1$ ,  $0 \leq y \leq 2$  and  $0 \leq z \leq 3$ . Compute the triple integral

$$\iiint_B (x + y + z) dx dy dz.$$

Since the limits of integration are constant we can perform the three integrals in any order. We will integrate over  $x, y, z$  in that order:

$$\int_0^3 \left( \int_0^2 \left( \int_0^1 (x + y + z) dx \right) dy \right) dz$$

<sup>2</sup><https://www.desmos.com/3d/1bcc69ab04>

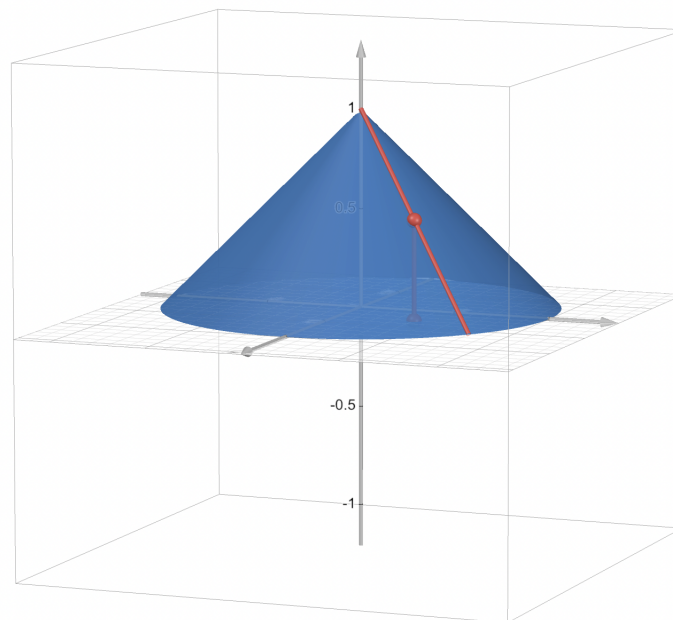
$$\begin{aligned}
&= \int_0^3 \left( \int_0^2 \left[ \frac{x^2}{2} + xy + xz \right]_{x=0}^{x=1} dy \right) dz \\
&= \int_0^3 \left( \int_0^2 \left( \frac{1}{2} + y + z \right) dy \right) dz \\
&= \int_0^3 \left[ \frac{1}{2}y + \frac{y^2}{2} + yz \right]_{y=0}^{y=2} dz \\
&= \int_0^3 (3 + 2z) dz \\
&= [3z + z^2]_{z=0}^{z=3} \\
&= 18.
\end{aligned}$$

I don't have anything interesting to say about this.

**Problem 5. Cylindrical Coordinates.** Consider a solid cone of radius 1 and height 1 whose base is the unit disk  $x^2 + y^2 \leq 1$  in the  $x, y$ -plane and whose vertex is at the point  $(0, 0, 1)$  in  $x, y, z$ -space.

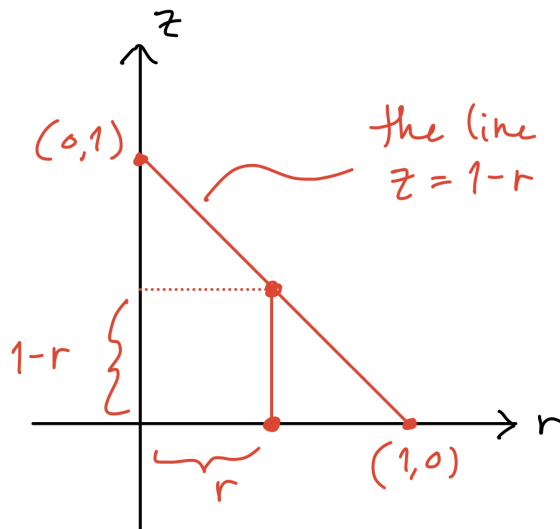
- Parametrize the cone using cylindrical coordinates:  $r, \theta, z$ .
- Compute the volume of the cone.
- Compute the center of mass  $(\bar{x}, \bar{y}, \bar{z})$ , assuming that the cone has constant density 1.  
[Hint: By symmetry we know that  $\bar{x} = 0$  and  $\bar{y} = 0$ , so you only have to compute  $\bar{z}$ .]

(a): We can parametrize the base circle by  $0 \leq r \leq 1$  and  $0 \leq \theta \leq 2\pi$ . Then for an arbitrary point in the base circle with coordinates  $r, \theta$  we must determine the maximum value of the  $z$ -coordinate.<sup>3</sup>



<sup>3</sup><https://www.desmos.com/3d/fe66f06a6f>

Since the cone is rotationally symmetric about the  $z$ -axis, the maximum value of  $z$  doesn't depend on  $\theta$ . To examine the relationship between  $z$  and  $r$ , consider a vertical slice through the origin:



Note that the surface of the cone intersects this slice in the straight line  $z = 1 - r$ . Hence we should take  $0 \leq z \leq 1 - r$ .

(b): Now we can use cylindrical coordinates to compute the volume of the cone:

$$\begin{aligned}
 \text{Vol}(\text{Cone}) &= \iiint_{\text{Cone}} 1 \, dx \, dy \, dz \\
 &= \iiint_{\text{Cone}} r \, dr \, d\theta \, dz && (dx \, dy \, dz = r \, dr \, d\theta \, dz) \\
 &= \int_0^{2\pi} 1 \, d\theta \cdot \int_0^1 r \cdot \left( \int_0^{1-r} 1 \, dz \right) dr && (\text{must integrate } z \text{ before } r) \\
 &= \int_0^{2\pi} 1 \, d\theta \cdot \int_0^1 r \cdot (1 - r) \, dr \\
 &= \int_0^{2\pi} 1 \, d\theta \cdot \int_0^1 (r - r^2) \, dr \\
 &= \int_0^{2\pi} 1 \, d\theta \cdot \left[ \frac{r^2}{2} - \frac{r^3}{3} \right]_0^1 \\
 &= 2\pi \cdot \left( \frac{1}{2} - \frac{1}{3} \right) \\
 &= \frac{\pi}{3}.
 \end{aligned}$$

Remark: The volume of a cone of radius  $r$  and height  $h$  is  $\pi r^2 h / 3$ . When  $r = 1$  and  $h = 1$  this formula gives  $\pi/3$ , which agrees with our computation.

(c): By symmetry we must have  $\bar{x} = 0$  and  $\bar{y} = 0$ . The  $z$  coordinate of the center of mass is

$$\bar{z} = \iiint_{\text{Cone}} z \, dx \, dy \, dz \, / \, \iiint_{\text{Cone}} 1 \, dx \, dy \, dz$$

$$\begin{aligned}
&= \frac{3}{\pi} \cdot \iiint_{\text{Cone}} z r \, dr \, d\theta \, dz \\
&= \frac{3}{\pi} \cdot \int_0^{2\pi} 1 \, d\theta \cdot \int_0^1 r \cdot \left( \int_0^{1-r} z \, dz \right) dr && \text{(must integrate } z \text{ before } r) \\
&= \frac{3}{\pi} \cdot \int_0^{2\pi} 1 \, d\theta \cdot \int_0^1 r \cdot (1-r)^2 / 2 \, dr \\
&= \frac{3}{2\pi} \cdot \int_0^{2\pi} 1 \, d\theta \cdot \int_0^1 (r^3 - 2r^2 + r) \, dr \\
&= \frac{3}{\pi} \cdot \int_0^{2\pi} 1 \, d\theta \cdot \left[ \frac{r^4}{4} - 2\frac{r^3}{3} + \frac{r^2}{2} \right]_0^1 \\
&= \frac{3}{2\pi} \cdot 2\pi \cdot \left( \frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right) \\
&= \frac{3}{2\pi} \cdot 2\pi \cdot \frac{1}{12} \\
&= \frac{1}{4}.
\end{aligned}$$

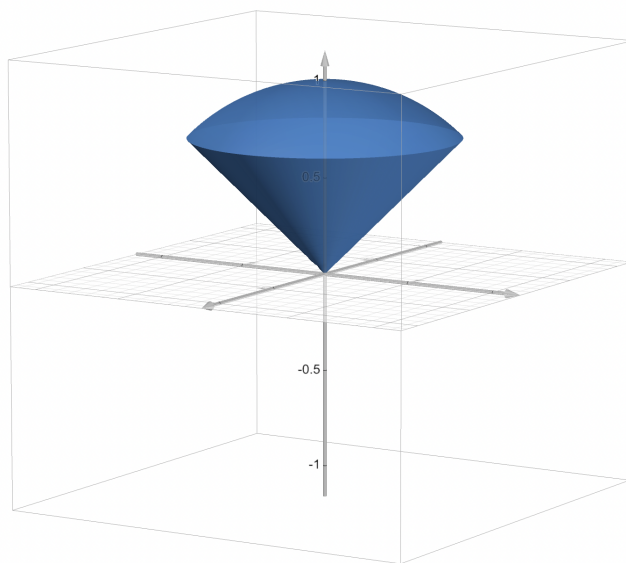
Hence the center of mass is  $(0, 0, 1/4)$ . Remark: We would get the same result for a cone of any radius and height. The center of mass is always exactly  $1/4$  of the way from the center of the base to the apex.

**Problem 6. Spherical Coordinates.** Consider the “ice-cream-cone-shaped” solid region  $E$  that is between the sphere  $x^2 + y^2 + z^2 = 1$  and the cone  $z^2 = x^2 + y^2$ , and satisfies  $z \geq 0$ . The volume is given by the triple integral:

$$\text{Vol}(E) = \iiint_E 1 \, dx \, dy \, dz.$$

Compute this integral by converting to spherical coordinates.

Here is a picture of the ice-cream cone, produced with Desmos<sup>4</sup>



<sup>4</sup><https://www.desmos.com/3d/8f0a6ad917>

This region is parametrized by  $0 \leq \rho \leq 1$ ,  $0 \leq \theta \leq 2\pi$  and  $0 \leq \varphi \leq \pi/4$ . (Note that the boundary of the cone has slope 1, hence angle  $\pi/4$  from the vertical. One can see this by intersecting the cone  $z^2 = x^2 + y^2$  with any vertical plane, such as  $y = 0$ , to get  $z^2 = x^2$ , and hence  $z = \pm x$ . This is a pair of lines of slope  $\pm 1$  in the  $x, z$ -plane.) Hence the volume is

$$\begin{aligned}
 \text{Vol}(E) &= \iiint_E 1 \, dx dy dz \\
 &= \iiint_E \rho^2 \sin \varphi \, d\rho d\theta d\varphi && (dx dy dz = \rho^2 \sin \varphi \, d\rho d\theta d\varphi) \\
 &= \int_0^1 \rho^2 \, d\rho \cdot \int_0^{2\pi} 1 \, d\theta \cdot \int_0^{\pi/4} \sin \varphi \, d\varphi && (\text{separable}) \\
 &= \left[ \frac{\rho^3}{3} \right]_0^1 \cdot [\theta]_0^{2\pi} \cdot [-\cos \varphi]_0^{\pi/4} \\
 &= \left( \frac{1}{3} \right) (2\pi) (-\cos(\pi/4) + \cos(0)) \\
 &= \frac{2\pi}{3} \left( -\frac{\sqrt{2}}{2} + 1 \right) \\
 &= \frac{2\pi}{3} \cdot \frac{2 - \sqrt{2}}{2} \\
 &= \frac{2\pi(2 - \sqrt{2})}{6} \\
 &\approx 0.6.
 \end{aligned}$$

Remark: I know that the most difficult part of this material is the visualization and parametrization of regions in  $\mathbb{R}^3$ . I strongly encourage you to use visualization tools such as Desmos and GeoGebra to build intuition:

<https://www.desmos.com/3d>

<https://www.geogebra.org/3d>

**Problem 1. Area of a Parametrized Region.** Given a region  $D$  in  $\mathbb{R}^2$ , the area is

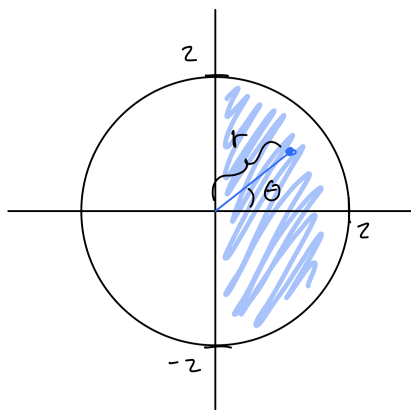
$$\text{Area}(D) = \iint_D 1 \, dx dy.$$

For each of the following problems you should (1) draw the region, (2) find a parametrization, (3) use your parametrization to compute the area.

- (a) The half-circle satisfying  $x^2 + y^2 \leq 4$  and  $x \geq 0$ . [Hint: Use polar coordinates.]
- (b) The region satisfying  $x^2 + y^2 \leq 4$  and  $x \geq 1$ . [Hint: Don't use polar coordinates. You will need the antiderivative

$$\int 2\sqrt{4-x^2} \, dx = x\sqrt{4-x^2} + 4 \arcsin(x/2).]$$

(a): In this case  $D$  is the right half of a circle of radius 2:

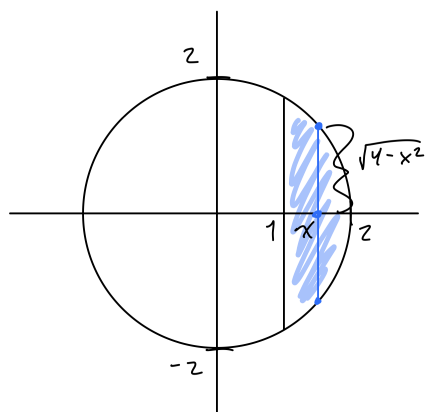


We parametrize  $D$  using polar coordinates with  $0 \leq r \leq 2$  and  $-\pi/2 \leq \theta \leq \pi/2$ . The area is

$$\begin{aligned} \iint 1 \, dx dy &= \iint r \, dr d\theta \\ &= \int_{-\pi/2}^{\pi/2} 1 \, d\theta \int_0^2 r \, dr \\ &= [\theta]_{\theta=-\pi/2}^{\theta=\pi/2} \left[ \frac{1}{2} r^2 \right]_{r=0}^{r=2} \\ &= \left[ \frac{\pi}{2} + \frac{\pi}{2} \right] \left[ \frac{1}{2} 2^2 \right] \\ &= 2\pi. \end{aligned}$$

Note that this agrees with the formula  $\pi(2)^2 = 4\pi$  for the area of the full circle.

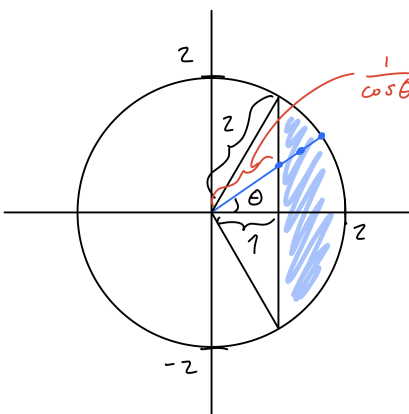
(b): Here is a picture of the region  $D$ :



We can parametrize this region by  $1 \leq x \leq 2$  and  $-\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}$ . The area is

$$\begin{aligned}
 \iint 1 \, dx \, dy &= \int_1^2 \left( \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 1 \, dy \right) dx \\
 &= \int_1^2 [y]_{y=-\sqrt{4-x^2}}^{y=\sqrt{4-x^2}} dx \\
 &= \int_1^2 2\sqrt{4-x^2} \, dx \\
 &= \left[ x\sqrt{4-x^2} + 4 \arcsin(x/2) \right]_{x=1}^{x=2} \\
 &= 2\sqrt{4-2^2} + 4 \arcsin(2/2) - 1\sqrt{4-1^2} - 4 \arcsin(1/2) \\
 &= 0 + 4 \arcsin(1) - \sqrt{3} - 4 \arcsin(1/2) \\
 &= 0 + 4(\pi/2) - \sqrt{3} - 4(\pi/6) \\
 &= 4\pi/3 - \sqrt{3}.
 \end{aligned}$$

Remark: It is also possible (but more difficult) to parametrize this region using polar coordinates. Consider the following picture:



The picture shows that  $-\pi/3 \leq \theta \leq \pi/3$  and  $1/\cos \theta \leq r \leq 2$ , so the area is

$$\iint 1 \, dx \, dy = \iint r \, dr \, d\theta$$

$$\begin{aligned}
&= \int_{-\pi/3}^{\pi/3} \left( \int_{1/\cos\theta}^2 r \, dr \right) d\theta \\
&= \int_{-\pi/3}^{\pi/3} \left[ \frac{1}{2} r^2 \right]_{r=1/\cos\theta}^{r=2} d\theta \\
&= \int_{-\pi/3}^{\pi/3} \left[ 2 - \frac{1}{2 \cos^2 \theta} \right] d\theta \\
&= \left[ 2\theta - \frac{\tan \theta}{2} \right]_{\theta=-\pi/3}^{\theta=\pi/3} \\
&= 2 \left[ \frac{\pi}{3} + \frac{\pi}{3} \right] - \frac{1}{2} [\tan(\pi/3) - \tan(-\pi/3)] \\
&= 2 \left[ \frac{2\pi}{3} \right] - \frac{1}{2} [\sqrt{3} + \sqrt{3}] \\
&= 4\pi/3 - \sqrt{3}.
\end{aligned}$$

The fact that we got the same answer each time means that the calculations are probably correct. This problem can also be solved without Calculus:

[https://en.wikipedia.org/wiki/Circular\\_segment#Arc\\_length\\_and\\_area](https://en.wikipedia.org/wiki/Circular_segment#Arc_length_and_area)

**Problem 2. Center of Mass of a 2D Region.** Let  $D$  be the region parametrized by  $0 \leq x \leq 2$  and  $x \leq y \leq 5x - 2x^2$ . Think of  $D$  as a solid with mass density 1.

- Compute the total mass  $M = \iint_D 1 \, dx dy$ .
- Compute the moments  $M_x = \iint_D x \, dx dy$  and  $M_y = \iint_D y \, dx dy$ .
- Compute the center of mass.
- Draw the region and its center of mass.

(a): The mass (i.e., the area) of the region  $D$  is

$$\begin{aligned}
M &= \iint_D 1 \, dx dy \\
&= \int_0^2 \left( \int_x^{5x-2x^2} 1 \, dy \right) dx \\
&= \int_0^2 (5x - 2x^2 - x) \, dx \\
&= \int_0^2 (4x - 2x^2) \, dx \\
&= \left[ 4 \frac{1}{2} x^2 - 2 \frac{1}{3} x^3 \right]_0^2 \\
&= 4 \frac{1}{2} 2^2 - 2 \frac{1}{3} 2^3 \\
&= 8/3.
\end{aligned}$$

(b): The moment in the  $x$  direction is

$$M_x = \iint_D x \, dx dy$$

$$\begin{aligned}
&= \int_0^2 \left( \int_x^{5x-2x^2} x \, dy \right) dx \\
&= \int_0^2 [xy]_{y=x}^{y=5x-2x^2} dx \\
&= \int_0^2 [x(5x-2x^2) - x^2] dx \\
&= \int_0^2 [4x^2 - 2x^3] dx \\
&= \left[ 4\frac{1}{3}x^3 - 2\frac{1}{4}x^4 \right]_0^2 \\
&= 4\frac{1}{3}2^3 - 2\frac{1}{4}2^4 \\
&= 8/3.
\end{aligned}$$

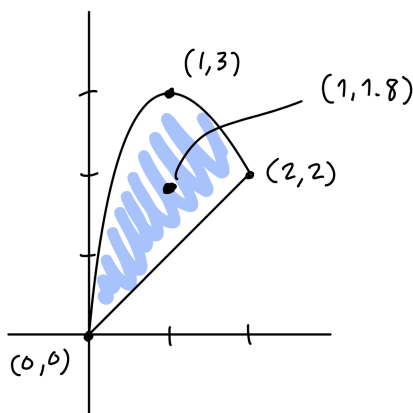
It is just a coincidence that  $M_x = M$ . The moment in the  $y$  direction is

$$\begin{aligned}
M_y &= \iint_D y \, dx \, dy \\
&= \int_0^2 \left( \int_x^{5x-2x^2} y \, dy \right) dx \\
&= \int_0^2 \left[ \frac{1}{2}y^2 \right]_{y=x}^{y=5x-2x^2} dx \\
&= \frac{1}{2} \int_0^2 [(5x-2x^2)^2 - x^2] dx \\
&= \frac{1}{2} \int_0^2 [4x^4 - 20x^3 + 24x^2] dx \\
&= \frac{1}{2} \left[ 4\frac{1}{5}x^5 - 20\frac{1}{4}x^4 + 24\frac{1}{3}x^3 \right]_0^2 \\
&= \frac{1}{2} \left( 4\frac{1}{5}2^5 - 20\frac{1}{4}2^4 + 24\frac{1}{3}2^3 \right) \\
&= 24/5.
\end{aligned}$$

(c): The center of mass is

$$(\bar{x}, \bar{y}) = \left( \frac{M_x}{M}, \frac{M_y}{M} \right) = \left( \frac{8/3}{8/3}, \frac{24/5}{8/3} \right) = \left( 1, \frac{9}{5} \right) = (1, 1.8).$$

(d): Here is a picture:



**Problem 3. Polar Coordinates.** Let  $x = r \cos \theta$  and  $y = r \sin \theta$ . We already know that

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \det \begin{pmatrix} x_r & x_\theta \\ y_r & y_\theta \end{pmatrix} = r.$$

The general theory predicts that we must also have

$$\frac{\partial(r, \theta)}{\partial(x, y)} = \det \begin{pmatrix} r_x & r_y \\ \theta_x & \theta_y \end{pmatrix} = \frac{1}{r}.$$

Check that this is true. [Hint:  $r = \sqrt{x^2 + y^2}$  and  $\theta = \arctan(y/x)$ .]

First we compute the derivatives using the one variable chain rule:

$$r_x = (1/2)(x^2 + y^2)^{-1/2}(2x) = x(x^2 + y^2)^{-1/2},$$

$$r_y = (1/2)(x^2 + y^2)^{-1/2}(2y) = y(x^2 + y^2)^{-1/2},$$

$$\theta_x = 1/(1 + (y/x)^2)(-y/x^2),$$

$$\theta_y = 1/(1 + (y/x)^2)(1/x).$$

The formulas for  $\theta_x$  and  $\theta_y$  can be simplified using  $1/(1 + (y/x)^2) = x^2/(x^2 + y^2)$  to get

$$\theta_x = x^2/(x^2 + y^2)(-y/x^2) = -y/(x^2 + y^2),$$

$$\theta_y = x^2/(x^2 + y^2)(1/x) = x/(x^2 + y^2).$$

Then we compute the determinant:

$$\begin{aligned} \det \begin{pmatrix} r_x & r_y \\ \theta_x & \theta_y \end{pmatrix} &= r_x \theta_y - r_y \theta_x \\ &= x(x^2 + y^2)^{-1/2} x/(x^2 + y^2) + y(x^2 + y^2)^{-1/2} y/(x^2 + y^2) \\ &= x^2(x^2 + y^2)^{-3/2} + y^2(x^2 + y^2)^{-3/2} \\ &= (x^2 + y^2)(x^2 + y^2)^{-3/2} \\ &= (x^2 + y^2)^{-1/2} \\ &= 1/\sqrt{x^2 + y^2} \\ &= 1/r. \end{aligned}$$

That was weirdly complicated, but we got the right answer.

**Problem 4. Center of Mass of a 3D Region.** Let  $D$  be the tetrahedron with vertices  $(0, 0, 0)$ ,  $(1, 0, 0)$ ,  $(0, 1, 0)$  and  $(0, 0, 1)$ . Think of  $D$  as a solid with constant mass density 1. This region can be parametrized by  $0 \leq x \leq 1$ ,  $0 \leq y \leq 1 - x$  and  $0 \leq z \leq 1 - x - y$ .

- (a) Compute the total mass  $M = \iiint_D 1 \, dx \, dy \, dz$ .  
 (b) Compute the moments

$$M_x = \iiint_D x \, dx \, dy \, dz, \quad M_y = \iiint_D y \, dx \, dy \, dz, \quad M_z = \iiint_D z \, dx \, dy \, dz.$$

[Hint: There might be a shortcut.]

- (c) Compute the center of mass.

(a): The total mass (i.e., the volume) is

$$\begin{aligned} M &= \iiint_D 1 \, dx \, dy \, dz \\ &= \int_0^1 \left( \int_0^{1-x} \left( \int_0^{1-x-y} 1 \, dz \right) dy \right) dx \\ &= \int_0^1 \left( \int_0^{1-x} (1 - x - y) dy \right) dx \\ &= \int_0^1 \left[ y - xy - \frac{1}{2}y^2 \right]_0^{1-x} dx \\ &= \int_0^1 \left[ (1-x) - x(1-x) - \frac{1}{2}(1-x)^2 \right] dx \\ &= \int_0^1 \left[ \frac{1}{2}x^2 - x + \frac{1}{2} \right] dx \\ &= \left[ \frac{1}{2} \frac{1}{3} x^3 - \frac{1}{2} x^2 + \frac{1}{2} x \right]_0^1 \\ &= \frac{1}{2} \frac{1}{3} - \frac{1}{2} + \frac{1}{2} \\ &= 1/6. \end{aligned}$$

(b): Because the shape  $D$  is symmetric under permuting  $x, y, z$  we know that  $M_x = M_y = M_z$ . It turns out that  $M_x$  is easiest to compute:

$$\begin{aligned} M_x &= \iiint_D x \, dx \, dy \, dz \\ &= \int_0^1 x \left( \int_0^{1-x} \left( \int_0^{1-x-y} 1 \, dz \right) dy \right) dx \\ &= \int_0^1 x \left( \int_0^{1-x} (1 - x - y) dy \right) dx \\ &= \int_0^1 x \left[ y - xy - \frac{1}{2}y^2 \right]_0^{1-x} dx \\ &= \int_0^1 x \left[ (1-x) - x(1-x) - \frac{1}{2}(1-x)^2 \right] dx \\ &= \int_0^1 x \left[ \frac{1}{2}x^2 - x + \frac{1}{2} \right] dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \left[ \frac{1}{2}x^3 - x^2 + \frac{1}{2}x \right] dx \\
&= \left[ \frac{1}{2} \frac{1}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2} \frac{1}{2}x^2 \right]_0^1 \\
&= \frac{1}{2} \frac{1}{4} - \frac{1}{3} + \frac{1}{2} \frac{1}{2} \\
&= 1/24.
\end{aligned}$$

(c): The center of mass is

$$(\bar{x}, \bar{y}, \bar{z}) = \left( \frac{M_x}{M}, \frac{M_y}{M}, \frac{M_z}{M} \right) = \left( \frac{1/24}{1/6}, \frac{1/24}{1/6}, \frac{1/24}{1/6} \right) = \left( \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right).$$

Remark: Consider the solid  $n$ -dimensional “simplex” with  $n + 1$  vertices:

$$(0, \dots, 0), (1, 0, \dots, 0), \dots, (0, \dots, 0, 1).$$

Using the same method, one can show that the  $n$ -dimensional “hypervolume” is  $1/n!$  and the center of mass is  $(1, 1, \dots, 1)/(n + 1)$ . However, as you can imagine, the computation is messy.

**Problem 5. Cylindrical Coordinates.** Let  $D$  be a solid cone of radius 1 and height 1. We can think of this as the solid region defined by  $x^2 + y^2 \leq 1$  and  $0 \leq z \leq 1 - \sqrt{x^2 + y^2}$ . Use cylindrical coordinates to compute the integral

$$\iiint_D z \, dx \, dy \, dz.$$

[Hint: Cylindrical coordinates are defined by  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = z$ , and satisfy  $\partial(x, y, z)/\partial(r, \theta, z) = r$ . That is,  $dx \, dy \, dz = r \, dr \, d\theta \, dz$ .]

In cylindrical coordinates, the cone  $D$  has parametrization  $0 \leq r \leq 1$ ,  $0 \leq \theta \leq 2\pi$  and  $0 \leq z \leq 1 - r$ . Hence

$$\begin{aligned}
\iiint_D z \, dr \, d\theta \, dz &= \iiint_D z r \, dr \, d\theta \, dz \\
&= 2\pi \int_0^1 r \left( \int_0^{1-r} z \, dz \right) dr \\
&= 2\pi \int_0^1 r \left[ \frac{1}{2} z^2 \right]_0^{1-r} dr \\
&= 2\pi \int_0^1 r(1-r)^2 dr \\
&= \pi \int_0^1 [r^3 - 2r^2 + r] dr \\
&= \pi \left[ \frac{1}{4} r^4 - 2 \frac{1}{3} r^3 + \frac{1}{2} r^2 \right]_0^1 \\
&= \pi \left( \frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right) \\
&= \pi/12.
\end{aligned}$$

Remark: If the cone has uniform density 1 then we just computed  $M_z$ . The volume of a cone is  $(1/3)\pi(\text{radius})^2(\text{height}) = \pi/3$  and by symmetry we have  $M_x = M_y = 0$ , hence the center of mass of the cone is

$$(\bar{x}, \bar{y}, \bar{z}) = \left( \frac{M_x}{M}, \frac{M_y}{M}, \frac{M_z}{M} \right) = \left( 0, 0, \frac{\pi/12}{\pi/3} \right) = \left( 0, 0, \frac{1}{4} \right).$$

That is, the center of mass on the main axis at  $1/4$  of the height. This same result holds for any cone of any radius and height.

**Problem 6.** Spherical coordinates  $\rho, \phi, \theta$  are defined by

$$x = \rho \sin \phi \cos \theta,$$

$$y = \rho \sin \phi \sin \theta,$$

$$z = \rho \cos \phi,$$

and satisfy  $\partial(x, y, z)/\partial(\rho, \phi, \theta) = \rho^2 \sin \phi$ . That is,  $dx dy dz = \rho^2 \sin \phi d\rho d\phi d\theta$ . Use spherical coordinates to compute the integral

$$\iiint_D \frac{1}{x^2 + y^2 + z^2} dx dy dz,$$

where  $D$  is the unit sphere. Even though the function  $f(x, y, z) = 1/(x^2 + y^2 + z^2)$  goes to infinity when  $(x, y, z) \rightarrow (0, 0, 0)$ , the integral is still finite.

$$\begin{aligned} \iiint_D \frac{1}{x^2 + y^2 + z^2} dx dy dz &= \iiint_D \frac{1}{\rho^2} \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \iiint_D \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} 1 d\theta \int_0^\pi 1 d\rho \int_0^\pi \sin \phi d\phi \\ &= 2\pi [-\cos \phi]_0^\pi \\ &= 2\pi [-\cos(\pi) + \cos(0)] \\ &= 2\pi [-(-1) + (1)] \\ &= 4\pi. \end{aligned}$$

Remark: This looked like the hardest problem on HW4, but it was actually the easiest!

Remark: The analogous integral in one dimension is  $\int_{-1}^1 (1/x^2) dx$ , which diverges. The analogous integral in two dimensions also diverges:

$$\begin{aligned} \iint_{\text{unit disk}} \frac{1}{x^2 + y^2} dx dy &= \iint \frac{1}{r^2} r dr d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 \frac{1}{r} dr \\ &= 2\pi [\ln(1) - \ln(0)] \\ &= 2\pi [0 - (-\infty)] \\ &= \infty. \end{aligned}$$

For some reason the three dimensional version converges. We will observe the same type of phenomenon when we study gravity.

## 2. Integration over a Rectangle.

Find the volume between the surface

$$z = (x+y)^2 = x^2 + 2xy + y^2 \text{ and the}$$

rectangle  $a_1 \leq x \leq a_2$  &  $b_1 \leq y \leq b_2$ .

$$\text{volume} = \iint \text{volume of skinny column}$$

$$= \iint \underbrace{f(x,y)}_{\text{height}} \underbrace{dx dy}_{\text{area of base}}$$

$$= \int_{b_1}^{b_2} \int_{a_1}^{a_2} f(x,y) dx dy$$

$$= \int_{b_1}^{b_2} \left( \int_{a_1}^{a_2} f(x,y) dx \right) dy$$

$$= \int_{b_1}^{b_2} \left( \int_{a_1}^{a_2} (x^2 + 2xy + y^2) dx \right) dy$$

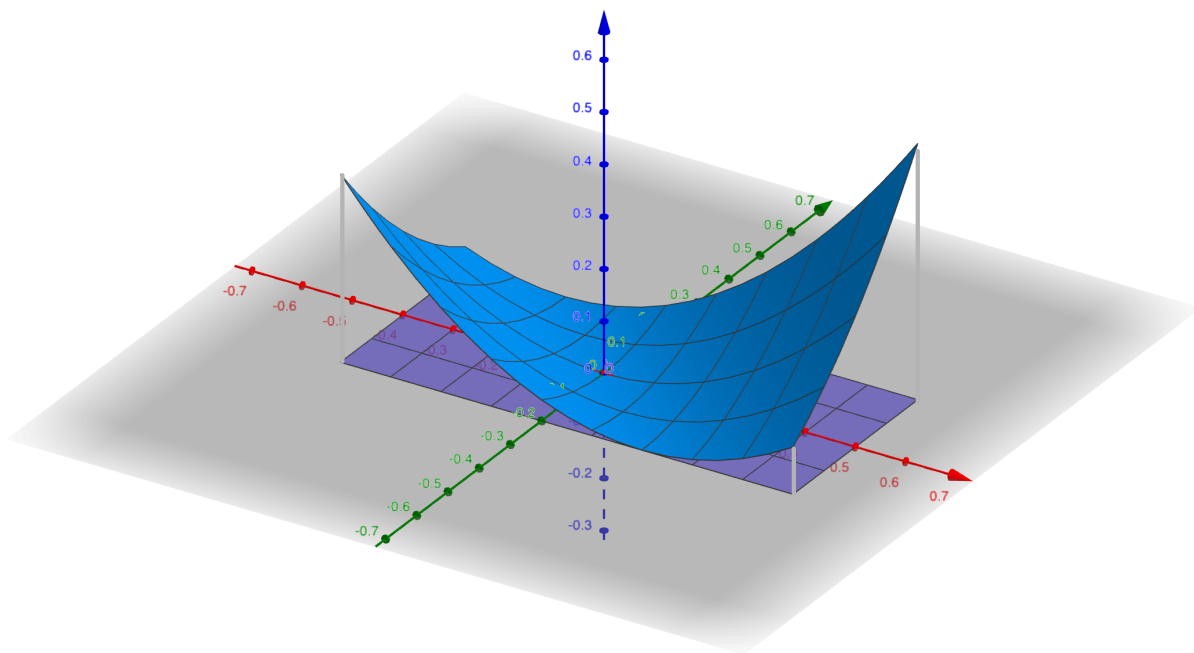
$$= \left( \frac{1}{3} (a_2^3 - a_1^3) y + \frac{1}{2} (a_2^2 - a_1^2) y^2 + \frac{1}{3} (a_2 - a_1) y^3 \right) \bigg|_{b_1}^{b_2}$$

$$\begin{aligned}
&= \frac{1}{3} (a_2^3 - a_1^3) (b_2 - b_1) \\
&\quad + \frac{1}{2} (a_2^2 - a_1^2) (b_2^2 - b_1^2) \\
&\quad + \frac{1}{3} (a_2 - a_1) (b_2^3 - b_1^3)
\end{aligned}$$

[We could factor out  $(a_2 - a_1)(b_2 - b_1)$  but that doesn't make it look nicer.]

Here is a picture of the bottom and top of the region when

$$(a_1, a_2, b_1, b_2) = (-0.4, 0.5, -0.2, 0.2)$$

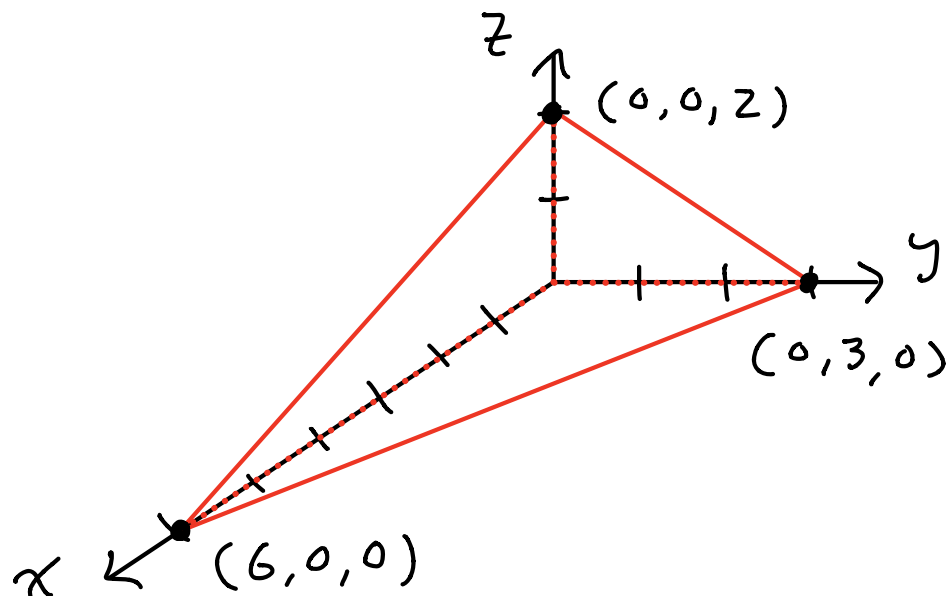


The volume of this region is 0.03

### 3. Integration over a Tetrahedron.

Consider the region in  $\mathbb{R}^3$  where

$$x, y, z \geq 0 \text{ \& \; } x + 2y + 3z \leq 6 :$$



Suppose this region has mass density

$$\rho(x, y, z) = 1 + x.$$

a) Compute the total mass.

I claim that we can parametrize the tetrahedron as follows :

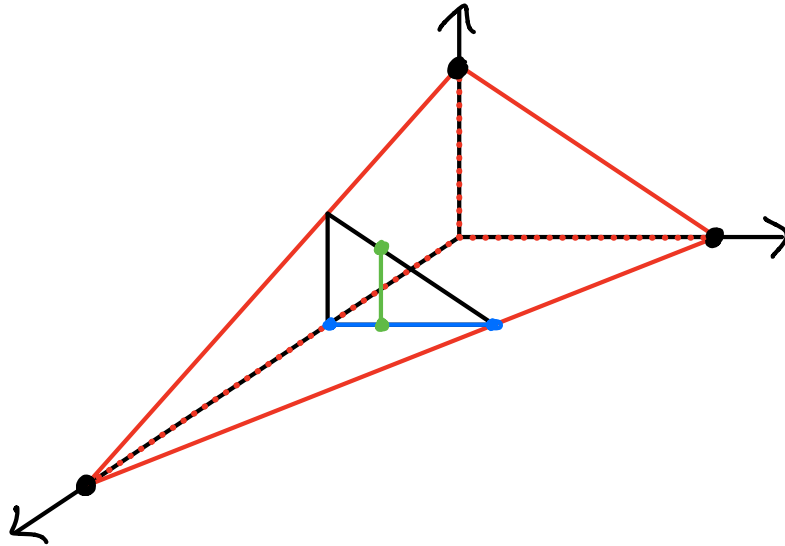
$$\begin{cases} 0 \leq x \leq 6 \\ 0 \leq y \leq 3 - 3x/6 \\ 0 \leq z \leq 2 - 2x/6 - 2y/3 \end{cases}$$

We could get this from the formula in the lecture notes with

$$(a, b, c) = (6, 3, 2).$$

Or we can do it from scratch.

- Fix some  $0 \leq x \leq 6$  and consider the triangular slice:



- The bottom of this triangular slice has  $z=0$ , so  $y$  satisfies

$$x + 2y + 0 \leq 6$$

$$2y \leq 6 - x$$

$$y \leq (6 - x)/2, \text{ as claimed } \checkmark$$

- Then for fixed  $x$  &  $y$ , the value of  $z$  must satisfy

$$x + 2y + 3z \leq 6$$

$$3z \leq 6 - 2y - x$$

$$z \leq (6 - 2y - x)/3, \text{ as claimed } \checkmark$$

Now we use this parametrization to compute the mass:

$$\text{mass} = \iiint \text{density} \cdot dV$$

$$= \iiint \rho(x, y, z) dx dy dz$$

$$= \iint \left( \int_0^{2-2y/3-x/3} (1+x) dz \right) dy dx$$

$$= \iint (1+x) \left( 2 - \frac{2}{3}y - \frac{1}{3}x \right) dy dx$$

$$= \int \left( \int_0^{3-x/2} \left[ (1+x) \left( 2 - \frac{1}{3}x \right) - \frac{2}{3}(1+x)y \right] dy \right) dx$$

$$= \int_0^6 \left[ (1+x) \left(2 - \frac{1}{2}x\right) \left(3 - \frac{x}{2}\right) - \frac{2}{3} (1+x) \frac{1}{2} \left(3 - \frac{x}{2}\right)^2 \right] dx$$

∴ computer

$$= \int_0^6 \left( 3 + 2x - \frac{11}{12}x^2 + \frac{1}{12}x^3 \right) dx$$

$$= 3 \cdot 6 + \frac{1}{2} 2 \cdot 6^2 - \frac{1}{3} \cdot \frac{11}{12} 6^3 + \frac{1}{4} \frac{1}{12} 6^4$$

$$= 15 \quad (\text{units of mass})$$

b) Part (a) was a lot of work.

This time I'll do everything on the computer, using the same parametrization of the domain.

$$M_{yz} = \iiint x(1+x) dV = \dots = 153/5$$

$$M_{xz} = \iiint y(1+x) dV = \dots = 99/10$$

$$M_{xy} = \iiint z(1+x) dV = \dots = 33/5.$$

c) So the center of mass is

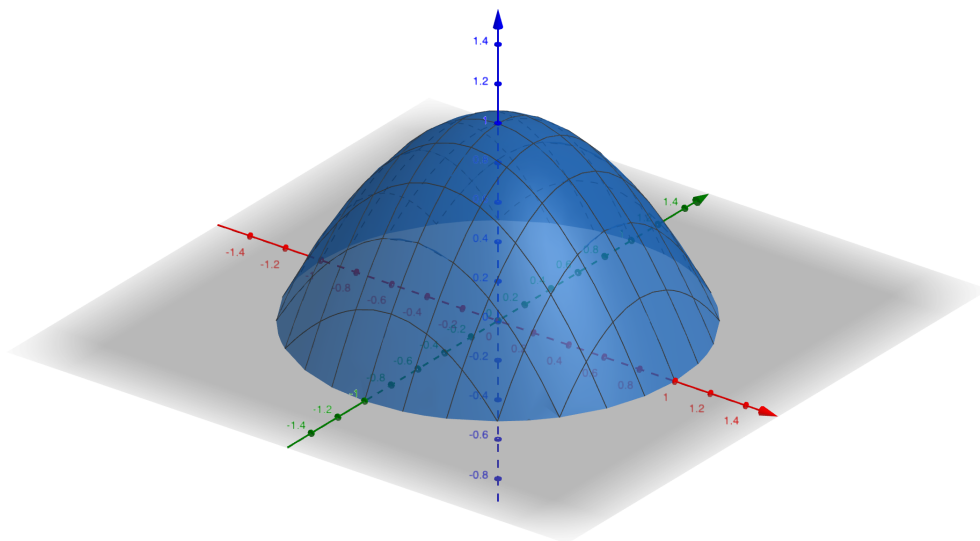
$$(\bar{x}, \bar{y}, \bar{z}) = \left( \frac{M_{yz}}{m}, \frac{M_{xz}}{m}, \frac{M_{xy}}{m} \right)$$

$$= \left( \frac{153/5}{15}, \frac{99/10}{15}, \frac{33/5}{15} \right)$$

$$= (2.04, 0.66, 0.44).$$

4. Cylindrical Coordinates.

Consider the parabolic dome between the unit disk  $x^2 + y^2 \leq 1$  and the surface  $z = 1 - x^2 - y^2$  :



a) We will compute the mass using cylindrical coordinates. Note that the dome is parametrized by

$$0 \leq r \leq 1$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq z \leq 1 - x^2 - y^2 = 1 - r^2,$$

so we get (density is 1)

$$\text{mass} = \iiint 1 dV$$

$$= \iiint r dr d\theta dz$$

$$= \iint \left( \int_0^{1-r^2} r dz \right) dr d\theta$$

$$= \iint r(1-r^2) dr d\theta$$

$$= \int \left( \int_0^1 (r - r^3) dr \right) d\theta$$

$$= \int \left( \frac{1}{2} r^2 - \frac{1}{4} r^4 \right) \Big|_0^1 d\theta$$

$$= \int_0^{2\pi} \frac{1}{4} d\theta$$

$$= 2\pi \cdot \frac{1}{4} = \pi/2.$$

[ We already did this computation in the lecture. ]

b) The moment about the  $xy$ -plane is

$$M_{xy} = \iiint z \, dV$$

$$= \iiint z \, r \, dr \, d\theta \, dz$$

$$= \iint \left( \int_0^{1-r^2} z \, r \, dz \right) dr \, d\theta$$

$$= \iint \left( \frac{1}{2} z^2 r \right) \Big|_0^{1-r^2} dr \, d\theta$$

$$= \int \left( \int_0^1 \frac{1}{2} (1-r^2)^2 r \, dr \right) d\theta$$

$$= \int \left( \int_0^1 \frac{1}{2} (r - 2r^3 + r^5) \, dr \right) d\theta$$

$$= \int_0^{2\pi} \frac{1}{2} \left( \frac{1}{2} r^2 - \frac{2}{4} r^4 + \frac{1}{6} r^6 \right) \Big|_0^1 d\theta$$

$$= \int_0^{2\pi} \frac{1}{12} d\theta$$

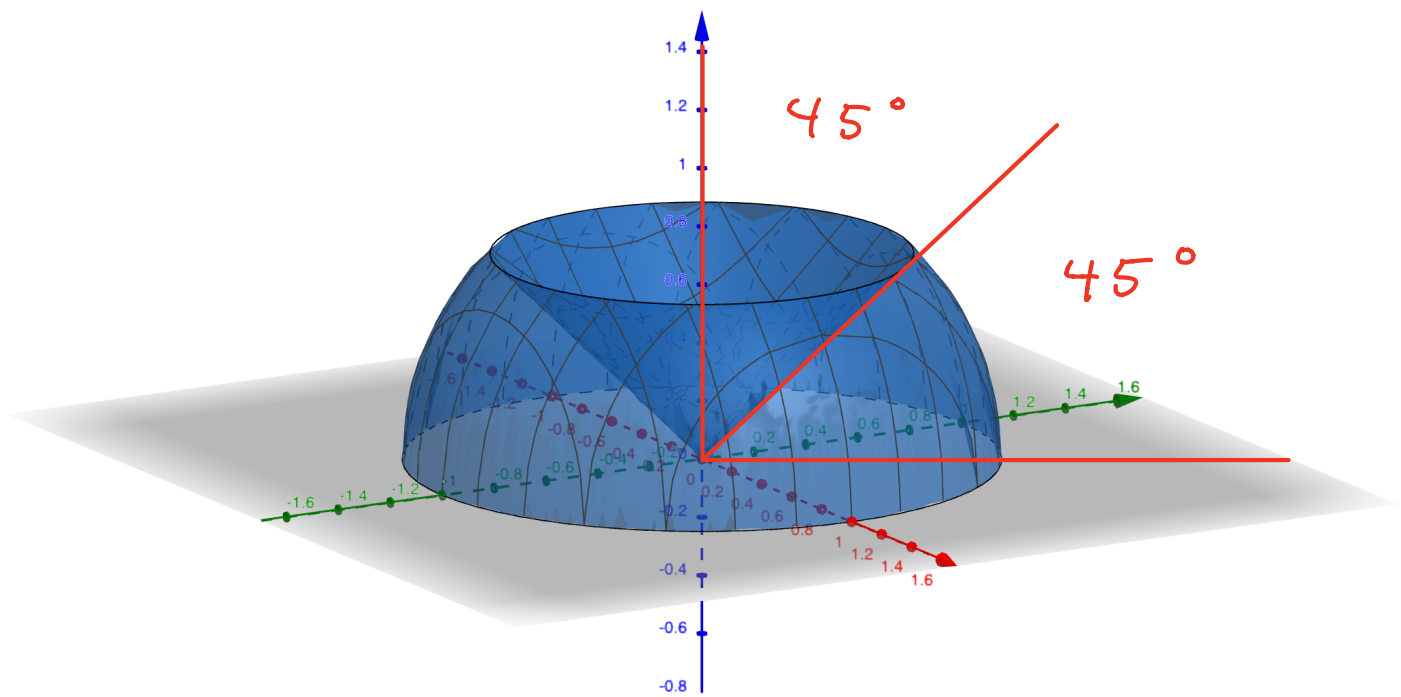
$$= 2\pi \cdot \frac{1}{12} = \pi/6.$$

c) By symmetry we have  $M_{xz} = M_{yz} = 0$ ,  
so the center of mass is

$$(\bar{x}, \bar{y}, \bar{z}) = \left( \frac{M_{yz}}{m}, \frac{M_{xz}}{m}, \frac{M_{xy}}{m} \right)$$

$$= \left( 0, 0, \frac{\pi/6}{\pi/2} \right) = \left( 0, 0, \frac{1}{3} \right).$$

6. We will compute the center of mass of the solid region above the  $xy$ -plane, below the cone  $z = \sqrt{x^2 + y^2}$  and inside the sphere  $x^2 + y^2 + z^2 \leq 1$ :



The region can be parametrized in spherical coordinates by

$$0 \leq \rho \leq 1$$

$$0 \leq \theta \leq 2\pi$$

$$\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{2}$$

a) The mass (or volume; we assume density is 1) is

$$m = \iiint dV$$

↙ integrand is "separable"

$$= \iiint \rho^2 \sin \varphi \, d\rho \, d\theta \, d\varphi$$

$$= \int_0^{2\pi} d\theta \int_0^1 \rho^2 \, d\rho \int_{\pi/4}^{\pi/2} \sin \varphi \, d\varphi$$

$$= 2\pi \cdot \frac{1}{3} \cdot \left( -\cos \frac{\pi}{2} + \cos \frac{\pi}{4} \right)$$

$$= 2\pi \cdot \frac{1}{3} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{3} \pi$$

b) The moment about the  $xy$ -plane is

$$M_{xy} = \iiint z \, dV$$

$$= \iiint \rho \cos \varphi \, \rho^2 \sin \varphi \, d\rho \, d\theta \, d\varphi$$

still "separable"

$$= \int_0^{2\pi} d\theta \int_0^1 \rho^3 d\rho \int_{\pi/4}^{\pi/2} \sin \varphi \cos \varphi d\varphi$$

$$= 2\pi \cdot \frac{1}{4} \int_{\pi/4}^{\pi/2} \sin \varphi \cos \varphi d\varphi$$

[Now we use a trig identity :

$$2 \sin \varphi \cos \varphi = \sin(2\varphi)$$

$$\sin \varphi \cos \varphi = \frac{1}{2} \sin(2\varphi)$$

]

$$= \cancel{2}\pi \cdot \frac{1}{4} \cdot \frac{1}{\cancel{2}} \int_{\pi/4}^{\pi/2} \sin(2\varphi) d\varphi$$

$$= \frac{\pi}{4} \left( -\frac{1}{2} \cos(2\varphi) \right)_{\pi/4}^{\pi/2}$$

$$= \frac{\pi}{4} \left( -\frac{1}{2} \overset{-1}{\cancel{\cos(\pi)}} + \frac{1}{2} \overset{0}{\cancel{\cos(\pi/2)}} \right)$$

$$= \pi/8$$

c) By symmetry we have  $M_{xz} = M_{yz} = 0$ ,  
so the center of mass is

$$(\bar{x}, \bar{y}, \bar{z}) = \left( \frac{M_{yz}}{m}, \frac{M_{xz}}{m}, \frac{M_{xy}}{m} \right)$$

$$= \left( 0, 0, \frac{\pi/8}{\pi\sqrt{2}/3} \right)$$

$$= (0, 0, 3\sqrt{2}/16)$$

$$\approx (0, 0, 0.265)$$

That seems reasonable.

Remark : If you throw this  
object in the air, its center  
of mass will trace out a  
parabolic curve.